

Rao-Blackwellised particle filtering for models with asynchronous observations

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1 Original Model Context

The original model (Guéant and Pu, 2019) describes complex latent Gaussian spatio-temporal dynamics and asynchronous convoluted observations of different spatial locations. The primary challenge in such continuous-time state-space models with discrete observation triggers is maintaining a tractable posterior distribution over the latent function when observations arrive asynchronously and independently across distinct coordinates.

2 Simplified Model

To analyse the Rao-Blackwellisation structure without the overhead of the full model, we define a simplified discrete-time correlated random walk in $\mathbb{R}^D$. Let $X_k \in \mathbb{R}^D$ denote the latent state at time step k . The state transition is governed by linear Gaussian dynamics:

$$X_k = X_{k-1} + \epsilon_k, \quad \epsilon_k \sim \mathcal{N}(0, Q), \quad (1)$$

where Q is a non-diagonal $D \times D$ covariance matrix inducing correlation across the coordinates.

Observations arrive asynchronously such that at each discrete time step k , only a single coordinate $d_k \in \{1, \dots, D\}$ is observed. For the sake of simplicity, and to allow for comparison with ground truth, we assume that the observation $y_k \in \mathbb{R}$ is conditionally Gaussian:

$$y_k = X_k^{d_k} + \nu_k, \quad \nu_k \sim \mathcal{N}(0, R), \quad (2)$$

where $X_k^{d_k}$ denotes the d_k -th element of the state vector X_k , and R is a scalar observation variance.

3 Inference Algorithms

3.1 Standard Particle Filter (PF)

A standard bootstrap particle filter approximates the posterior $p(X_{0:k} \mid y_{1:k})$ via a set of N weighted particles $\{X_k^{(i)}, w_k^{(i)}\}_{i=1}^N$.

1. **Prediction:** Sample $X_k^{(i)} \sim \mathcal{N}(X_{k-1}^{(i)}, Q)$ for $i = 1, \dots, N$.
2. **Update:** Compute unnormalized log-weights based on the 1D observation:

$$\log \tilde{w}_k^{(i)} = \log w_{k-1}^{(i)} + \log \mathcal{N}(y_k \mid [X_k^{(i)}]_{d_k}, R). \quad (3)$$

Normalize weights $w_k^{(i)}$.

3. **Resampling:** Perform resampling on the particles $\{X_k^{(i)}\}$ according to weights $\{w_k^{(i)}\}$ at every step, subsequently resetting $w_k^{(i)} = 1/N$.

3.2 Rao-Blackwellized Particle Filter (RB-PF)

The RB-PF targets the marginal posterior of the sequence of sampled coordinates and recovers the complementary in closed form. Let $U_k = X_k^{d_k}$ be the realization of the coordinate observed at time k . The posterior factorizes as:

$$p(X_{0:k} \mid y_{1:k}) = p(X_{0:k}^{-d} \mid U_{1:k})p(U_{1:k} \mid y_{1:k}). \quad (4)$$

The particle filter approximates the augmented distribution $p(U_{1:k} \mid y_{1:k})$ using weighted samples $\{U_{1:k}^{(i)}, w_k^{(i)}\}_{i=1}^N$. Conditional on a specific trajectory $U_{1:k}^{(i)}$, the remaining state dimensions are analytically tracked via a Gaussian distribution $\mathcal{N}(\mu_k^{(i)}, P_k)$. Because the dynamics are linear and the observed coordinates are identical across particles, the covariance matrix P_k is deterministic and particle-independent. Under a bootstrap proposal, the algorithm takes the following form.

1. **Prediction (Prior over full state):** Compute the predictive mean and covariance for the full state:

$$\mu_{k|k-1}^{(i)} = \mu_{k-1}^{(i)}, \quad P_{k|k-1} = P_{k-1} + Q. \quad (5)$$

Extract prior statistics to form the predictive distribution for the specific observed coordinate d_k , $p(U_k \mid U_{1:k-1}^{(i)}) = \mathcal{N}(m_k^{(i)}, s_k)$, where $m_k^{(i)} = [\mu_{k|k-1}^{(i)}]_{d_k}$ and $s_k = [P_{k|k-1}]_{d_k, d_k}$.

2. **Bootstrap Proposal Sampling:** Sample the 1D coordinate realization $u_k^{(i)}$ from its predictive prior:

$$u_k^{(i)} \sim \mathcal{N}(m_k^{(i)}, s_k). \quad (6)$$

3. **Weight Update:** Compute unnormalized log-weights based on the 1D observation of the sampled coordinate, identical to the standard particle filter:

$$\log \tilde{w}_k^{(i)} = \log w_{k-1}^{(i)} + \log \mathcal{N}(y_k \mid u_k^{(i)}, R). \quad (7)$$

Normalize weights $w_k^{(i)}$ in log-space.

4. **Exact Marginalization:** Condition the full state on the specific realization $u_k^{(i)}$. By treating $u_k^{(i)}$ as a noise-free measurement of $X_k^{d_k}$, we can apply the standard Kalman update. The gain is $K = \frac{1}{s_k} P_{k|k-1, (:, d_k)}$.

$$\mu_k^{(i)} = \mu_{k|k-1}^{(i)} + K \left(u_k^{(i)} - m_k^{(i)} \right) \quad (8)$$

$$P_k = P_{k|k-1} - K [P_{k|k-1, (d_k, :)}] \quad (9)$$

Note: This operation forces the d_k -th element of $\mu_k^{(i)}$ to equal exactly $u_k^{(i)}$. Correspondingly, the d_k -th row and column of P_k collapse to exactly zero, reflecting no residual uncertainty in the value of the sampled coordinate.

5. **Resampling:** Perform resampling on the conditional statistics $\{\mu_k^{(i)}\}$ based on $\{w_k^{(i)}\}$, subsequently resetting $w_k^{(i)} = 1/N$.

3.3 Compute-Efficient Lazy Evaluation

In high-dimensional spaces, updating the full state mean vector $\mu_k^{(i)}$ eagerly at every observation step requires $\mathcal{O}(ND)$ operations. An exact lazy evaluation strategy defers these updates until a coordinate is explicitly queried, preserving exactness by conditioning on all intermediate coordinates sampled since the last update.

Let $\tau(d)$ denote the time index when coordinate $d \in \{1, \dots, D\}$ was last observed. Because the latent dynamics are linear and Gaussian, the predictive covariance matrix P_k and the Kalman gain vectors K_k are deterministic and independent of the particle realizations.

For each particle i , the sequence of sampled innovations $\Delta_t^{(i)} = u_t^{(i)} - m_t^{(i)}$ is retained. When coordinate d_k is observed at time k , its exact conditional prior mean $m_k^{(i)}$ is computed by accumulating the historical innovations:

$$m_k^{(i)} = \mu_{\tau(d_k)}^{(i), d_k} + \sum_{t=\tau(d_k)+1}^{k-1} K_t^{d_k} \Delta_t^{(i)} \quad (10)$$

The prior variance is the diagonal element $s_k^{(i)} = [P_{k|k-1}]_{d_k, d_k}$ from the globally tracked covariance. Following the sampling of $u_k^{(i)}$, the innovation

$\Delta_k^{(i)}$ is stored, the tracked mean is updated for the queried dimension to $\mu_k^{(i), d_k} = u_k^{(i)}$, and the time index advances via $\tau(d_k) = k$. This formulation is analytically identical to the standard RB update.

3.4 Backward Sampling (Smoothing)

To obtain the smoothing distribution $p(X_{0:T} | y_{1:T})$, we apply Forward Filtering Backward Simulation (FFBSi) to the Rao-Blackwellized particles. The forward filter approximates the distribution at time k as a Gaussian mixture $\sum_{i=1}^N w_k^{(i)} \mathcal{N}(\mu_k^{(i)}, P_k)$. Because the observation model and timings are identical across particles, the state covariance P_k is deterministic and independent of the particle index i .

1. **Initialization:** At the terminal time T , sample a particle index I_T with probability proportional to the final filtering weights $w_T^{(i)}$. Draw the terminal full state X_T^* from the corresponding conditional Gaussian:

$$X_T^* \sim \mathcal{N}(\mu_T^{(I_T)}, P_T). \quad (11)$$

2. **Backward Simulation:** For $k = T - 1$ down to 0, the smoothed density given the future sampled trajectory state X_{k+1}^* is derived via Bayes' theorem:

$$p(X_k | X_{k+1}^*, y_{1:k}) \propto p(X_{k+1}^* | X_k) \sum_{i=1}^N w_k^{(i)} \mathcal{N}(X_k | \mu_k^{(i)}, P_k). \quad (12)$$

Because the latent transition $p(X_{k+1}^* | X_k) = \mathcal{N}(X_{k+1}^* | X_k, Q)$ is linear Gaussian, the product yields a new Gaussian mixture:

$$p(X_k | X_{k+1}^*, y_{1:k}) \approx \sum_{i=1}^N w_{k|k+1}^{(i)} \mathcal{N}(X_k | m_{k|k+1}^{(i)}, \Sigma_{k|k+1}). \quad (13)$$

The backward mixing weights are obtained by evaluating the predictive density of X_{k+1}^* :

$$w_{k|k+1}^{(i)} \propto w_k^{(i)} \mathcal{N}(X_{k+1}^* | \mu_k^{(i)}, P_k + Q), \quad (14)$$

where the weights $\{w_{k|k+1}^{(i)}\}_{i=1}^N$ are subsequently normalized. The conditional mean and covariance follow standard Rauch-Tung-Striebel (RTS) smoothing equations. Given the deterministic smoother gain $J_k = P_k(P_k + Q)^{-1}$, these are computed as:

$$m_{k|k+1}^{(i)} = \mu_k^{(i)} + J_k (X_{k+1}^* - \mu_k^{(i)}), \quad (15)$$

$$\Sigma_{k|k+1} = P_k - J_k P_k. \quad (16)$$

Sample an index I_k according to the backward weights $w_{k|k+1}^{(i)}$, and simulate the state:

$$X_k^* \sim \mathcal{N}\left(m_{k|k+1}^{(I_k)}, \Sigma_{k|k+1}\right). \quad (17)$$

Repeating this backward pass multiple times produces independent realizations from the joint smoothing distribution. The structural independence of P_k , $P_k + Q$, and J_k from the particle indices ensures the backward pass can be vectorized efficiently alongside the lazy evaluation scheme.

4 Simulation Study

We evaluate the standard PF and the RB-PF for $N = 10\,000$ on a $D = 10$ -dimensional system over 150 time steps, executing systematic resampling unconditionally at every iteration. Ground truth inference is obtained via an exact Kalman Filter. Both filters use the standard bootstrap proposal mechanism, isolating Rao-Blackwellisation as the only point of algorithmic divergence. Results are illustrated in Table 1 and Figure 1 and code to reproduce is here. Additionally, the lazy implementation was ~ 3.5 times more efficient on this example.

Table 1: Root Mean Square Error (RMSE) against ground truth trajectory and exact inference.

Filter	RMSE (vs True State)	RMSE (vs Kalman Filter)
Kalman Filter (Exact)	6.8645	—
Standard PF	7.8682	2.8071
RBPF	6.8550	0.1091

References

Guéant, O. and Pu, J. (2019). Mid-price estimation for european corporate bonds: a particle filtering approach.

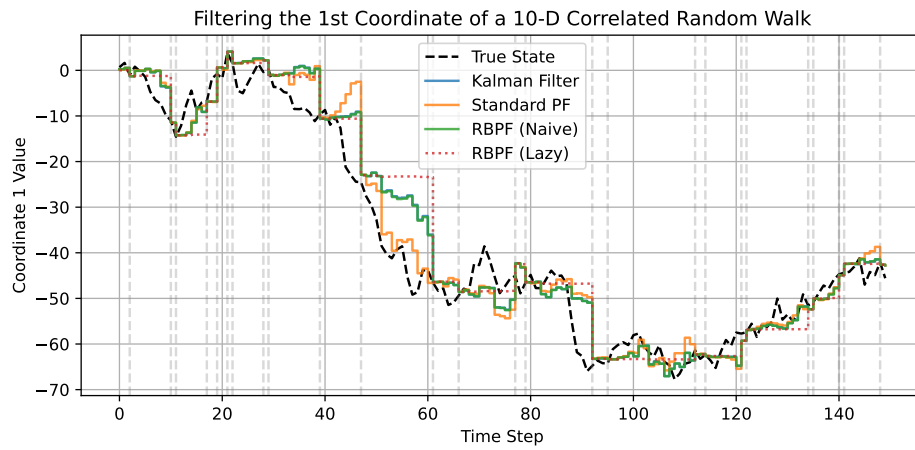


Figure 1: First component mean estimates for the different methods. Vertical dashed lines correspond to observation times for the first component.